Linux Task 02-Anju: Users, Groups & File Permissions

Objective :

The purpose of this task is to understand Linux user management, groups, ownership, and file permissions.

1. What is your current username?

My current username is **anju_sahana**.
A username is used to identify a user account in the Linux system.
It helps the operating system know who is currently logged in.

The whoami command is used to display the current username.
In my system, the username is **anju_sahana**.

2. What is UID?

UID stands for **User ID**.
It is a unique number assigned to each user in Linux.
The operating system uses the UID to identify users internally.
From the id command output, my UID is **1*****.
Regular users usually have a UID starting from 1***.

3. What is GID?

GID stands for **Group ID**.
It is a unique number assigned to a user group in Linux.
Users belonging to the same group can share permissions and resources.
From the id command output, my GID is **1*****.

4. What information does /etc/passwd contain?

The /etc/passwd file contains information about all user accounts in the Linux system.
It stores details such as the username, UID, GID, home directory, and default shell.
Each line in the file represents one user account.
System users and normal users are both listed in this file.

Part B: Create Users & Groups

1. Create Groups

sudo groupadd interns
sudo groupadd cyberteam

Purpose

- groupadd interns creates a group named interns.
- groupadd cyberteam creates a group named cyberteam.

2. Create Users

sudo useradd -m student1

sudo useradd -m student2

sudo useradd -m student3

Purpose

- useradd -m creates a new user and a home directory.

3. Add Users to Groups

sudo usermod -aG interns student1

sudo usermod -aG interns student2

sudo usermod -aG cyberteam student2

sudo usermod -aG cyberteam student3

Group Membership

User	Group(s)
student1	interns
student2	interns, cyberteam
student3	cyberteam

Verify

groups - anju_sahana adm cdrom sudo dip plugdev users

id student1 - id student1

uid=1001(student1) gid=1003(student1) groups=1003(student1),1001(interns),1002(cyberteam)

Part C: File Ownership

Create the Project Folder

Run: mkdir CyberSecurity_Project

Move into the Folder

cd CyberSecurity_Project

Create the Files

touch report.txt notes.txt credentials.txt

Check Ownership – ls -l

-rw-r--r-- 1 anju_sahana anju_sahana 0 Jun 12 report.txt

-rw-r--r-- 1 anju_sahana anju_sahana 0 Jun 12 notes.txt

-rw-r--r-- 1 anju_sahana anju_sahana 0 Jun 12 credentials.txt

Change Ownership

sudo chown student1 credentials.txt

-rw-r--r-- 1 anju_sahana anju_sahana 0 Jun 12 report.txt

-rw-r--r-- 1 anju_sahana anju_sahana 0 Jun 12 notes.txt

-rw-r--r-- 1 student1 anju_sahana 0 Jun 12 credentials.txt

Item	Value
Folder Name	CyberSecurity_Project
Files Created	report.txt, notes.txt, credentials.txt
Original Owner	anju_sahana
New Owner	student1
File Changed	credentials.txt
Command Used	sudo chown student1 credentials.txt

Part D: File Permissions

Create the File - touch security_policy.txt

Verify - ls -l security_policy.txt

Set Read Only Permission (r--r--r--) - chmod 444 security_policy.txt

Verify - ls -l security_policy.txt

Set Read & Write Permission (rw-rw-rw-) - chmod 666 security_policy.txt

Verify - ls -l security_policy.txt

Set Full Access (rwxrwxrwx) - chmod 777 security_policy.txt

Verify - ls -l security_policy.txt

Permission Type	Symbolic Form	Command
Read Only	r--r--r--	chmod 444 security_policy.txt
Read & Write	rw-rw-rw-	chmod 666 security_policy.txt

Full Access	rwxrwxrwx	chmod 777 security_policy.txt

Part E: Permission Analysis

Permission	Owner Rights	Group Rights	Other User Rights	Real-World Use Case
755	Read, Write, Execute	Read, Execute	Read, Execute	Scripts, websites, and application folders
644	Read, Write	Read	Read	Normal text files and configuration files
777	Read, Write, Execute	Read, Write, Execute	Read, Write, Execute	Testing environments (not recommended for security)
600	Read, Write	No Access	No Access	Password files and private documents

700	Read, Write, Execute	No Access	No Access	Personal scripts and private directories

Explanation

755

The owner can read, write, and execute the file. Group members and others can only read and execute it. Commonly used for executable scripts and folders.

644

The owner can read and modify the file. Group members and others can only read it. Commonly used for documents and configuration files.

777

Everyone has full control over the file. Any user can read, modify, or execute it. It is generally avoided because it creates security risks.

600

Only the owner can read and write the file. No one else can access it. Suitable for confidential files and passwords.

700

Only the owner has full access. Group members and others cannot access the file or directory. Used for private folders and personal scripts.